

[Total No. of Printed Pages : 2

AU-5002-CBGS

Examination, December 2020

Measurement And Control

Maximum Marks : 70

1.
 - a) Describe with examples the applications of measurement systems.
 - b) Explain about the following static characteristics:
 - i) Accuracy
 - ii) Sensitivity
 - iii) Range
 - iv) Precision
2.
 - a) Discuss various modes of presenting data.
 - b) Discuss accuracy and precision.
3.
 - a) Explain the construction details and working principle of bimetallic thermometer with neat sketch.
 - b) Describe the construction and working of a Bourdon tube. Describe the C type, spiral type and helical type Bourdon gauges with neat diagrams.
4.
 - a) What is uncertainty analysis? How it is done?
 - b) Write about data acquisition and data reduction.

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[2]

5. Discuss with neat sketches following measuring devices:-
 - a) Bimetallic thermometer
 - b) Thermistors
 - c) Total and static pressure measurement in moving fluids.
 - d) Rotameter
6. Describe the bellows pressure gauge with the help of neat figure.
7.
 - a) Discuss the principle of engine testing dynamometers.
 - b) Explain in brief the transfer functions of elements, system and processes.
8. Write short notes on the followings:
 - a) Principle of working of thermocouple
 - b) Transfer function
 - c) Piezo-electric load cells
